Patient-Level Spectral Classification with Generative Augmentation Results and Discussion

1 Results and Discussion

1.1 Evaluation protocol and interpretation of the metrics

The experimental comparison comprised two nested selection stages followed by a patient-level outer evaluation. Stage 1 screened recent generative augmentation approaches with a fixed Patch Transformer probe. Stage 2 jointly evaluated the classifier and the spectral representation after applying the generator chosen from the training patients of each outer fold. Final predictions were obtained exclusively for original held-out patients; no synthetically generated spectrum was treated as an outer-test observation. The results therefore distinguish the inner-CV scores used to select configurations from the outer-fold estimates used to discuss generalization. Importantly, the values marked *Selected* in the first two tables count choices made by the inner-CV procedure across outer folds; they are not numbers of correctly classified patients.

The supplied outer-evaluation tables contain one prediction for each of 27 patients, of whom 20 were sick (positive class) and seven were healthy (negative class). Because these class sizes differ, balanced accuracy (BA) is used as the principal classification metric:

$$\mathrm{BA} = \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} + \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}\right) = \frac{\mathrm{Sensitivity} + \mathrm{Specificity}}{2}. \tag{1}$$

Accuracy, area under the receiver-operating-characteristic curve (AUC), average precision (AP), F1 score, and Matthews correlation coefficient (MCC) provide complementary information. Throughout, differences in BA are expressed in percentage points (pp), rather than as relative percentage changes. Descriptive inner-CV scores and point estimates from the limited outer sample are not treated as evidence of statistically established improvement.

1.2 Stage 1: Effect of the augmentation method and synthesis budget

Table 1 and Fig. 1 summarize the fixed-probe analysis. With real training spectra alone, the Patch Transformer obtained a mean inner-CV BA of 70.4% with a 9.8 pp standard deviation. Rectified Flow with 12 synthetic spectra per class achieved the largest mean BA, 72.7% (SD 7.6 pp), and was selected in five of the 27 outer folds. The figure displays a gain of approximately 2.4 pp, while subtraction of the rounded mean BA values in the table gives 2.3 pp; both describe the same small observed advantage at the precision shown. Stochastic Interpolant with 30 spectra per class reached 72.1%, VP Flow Matching with 12 per class reached 71.9%, and Latent EDM with 30 per class reached 71.8%. All were above the 70.4% real-only mean, although the difference between candidates was small relative to the fold-to-fold variation.

The synthesis budget did not have a uniform effect. Increasing Rectified Flow from 12 to 30 samples per class reduced its mean BA from 72.7% to 71.4%. VP Flow Matching also decreased from 71.9% to 71.3%. In contrast, Stochastic Interpolant rose from 70.1% to 72.1%, and Latent EDM rose from 69.0% to 71.8%. The largest deterioration relative to real-only training occurred for Latent EDM with 12 samples per class (69.0%, a decrease of 1.4 pp). These contrasts indicate that adding more artificial observations is not intrinsically beneficial and that sample count must be considered jointly with generator choice.

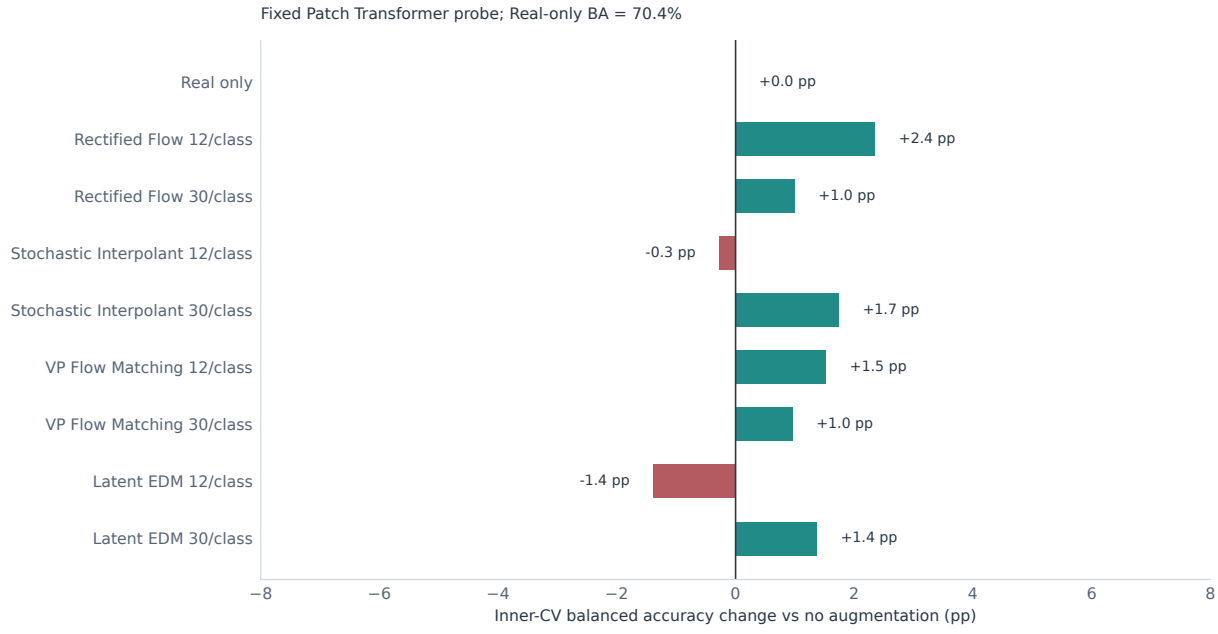
The corresponding pooled inner-CV AUCs were 0.751 for real-only training, 0.773 for Rectified Flow with 12 samples per class, and 0.775 for Stochastic Interpolant with 30 samples per class. Thus, the generator maximizing mean BA was not the same as the generator with the highest reported pooled AUC. Moreover, both Rectified Flow with 12 samples per class and real-only training were selected in five folds, compared with four for Rectified Flow with 30 samples per class. No generator was consistently preferred across all outer training partitions. These results motivate fold-specific augmentation selection rather than a global claim that one generative architecture dominates the data.

Table 1: Stage 1: generator screening with a fixed Patch Transformer probe, based only on the training patients within each outer fold. Mean and standard deviation (SD) refer to inner-CV balanced accuracy. AUC denotes the pooled inner-CV area under the ROC curve.

Candidate	Mean BA (%)	SD (pp)	AUC	Selected	Time (s)
Rectified Flow 12/class	72.7	7.6	0.773	5	0.612
Stochastic Interpolant 30/class	72.1	8.2	0.775	3	1.034
VP Flow Matching 12/class	71.9	8.1	0.773	2	0.605
Latent EDM 30/class	71.8	10.5	0.759	3	1.035
Rectified Flow 30/class	71.4	8.8	0.770	4	1.020
VP Flow Matching 30/class	71.3	7.8	0.774	2	1.034
Real only	70.4	9.8	0.751	5	0.225
Stochastic Interpolant 12/class	70.1	8.9	0.773	2	0.605
Latent EDM 12/class	69.0	7.6	0.740	1	0.606

Note: Every candidate was screened across 27 outer folds; *Selected* is the number of outer folds in which the corresponding option was preferred by inner CV, not an outer-test success count. Time is the reported mean cold-start training time per inner fold.

Figure 1 | Which generator helps the fixed classification probe?



Positive = better than the real-only probe. Selection uses training patients only. These are descriptive INNER-CV results, not outer-test improvements.

Figure 1: Inner-CV balanced-accuracy differences relative to the fixed real-only probe for each generator and synthesis budget. Positive bars favor augmentation; these are screening scores, not outer-test effects.

1.3 Stage 2: Joint classifier and spectral representation ablation

The second stage tested four neural classifiers—Patch Transformer, Spectral Mixer, ModernTCN-lite, and ConvNeXt-1D—with five spectral representations: raw standard-normal-variate (SNV) spectra, multiscale patches, gradient-plus-shape information, scale-space bands, and a frequency hybrid. The complete 20-combination ablation is reported in Table 2, with its performance matrix in Fig. 2. Generator configuration was inherited from the first-stage selection within each outer training fold.

The highest mean inner-CV BA was obtained by Patch Transformer with raw SNV spectra (80.8%, SD 6.7 pp), with a pooled inner AUC of 0.818. Spectral Mixer with raw SNV spectra ranked second for BA (79.8%, SD 10.8 pp), but attained a higher reported pooled inner AUC of 0.865. Spectral Mixer with multiscale patches achieved 79.5% BA and the highest AUC in the ablation (0.884). The differences between the top mean BA values are smaller than their reported cross-fold SDs and should not be interpreted as statistically resolved rankings.

Selection frequency revealed a different pattern from the ordering of mean scores. Spectral Mixer with raw SNV was chosen in nine outer folds, Spectral Mixer with multiscale patches in four, and Patch Transformer with raw SNV in four. The remaining selected configurations were distributed across other model–representation pairs. A candidate selected more often was not necessarily the single candidate with the highest mean score, reflecting variation in the data partitions used for inner selection. The counts do not establish performance superiority on the outer held-out patients.

The representation effects depended on the model architecture. For the Patch Transformer, raw SNV (80.8%) substantially exceeded multiscale patches (71.1%) and the frequency hybrid (71.5%), while gradient-plus-shape features reached 75.9%. Spectral Mixer was comparatively robust across raw SNV (79.8%), multiscale patches (79.5%), and scale-space bands (78.5%). Conversely, gradient-plus-shape gave the best mean BA for ModernTCN-lite (75.0%) and ConvNeXt-1D (74.9%). These interactions show why evaluating a feature transformation independently of the classifier can obscure the best-performing combination.

The frequency hybrid did not provide the highest mean BA for any classifier. Furthermore, neither feature engineering nor additional spectral decomposition consistently improved upon the raw SNV input. This observation should be interpreted for the evaluated configurations only; it does not show that spectral feature engineering is generally unnecessary.

Table 2: Stage 2: classifier–representation combinations after fold-specific generator selection. All scores are based on inner CV; selection counts indicate how often a pair was selected across 27 outer folds.

Classifier + representation	Mean BA (%)	SD (pp)	AUC	Selected	Time (s)
Patch Transformer + SNV raw	80.8	6.7	0.818	4	0.742
Spectral Mixer + SNV raw	79.8	10.8	0.865	9	0.342
Spectral Mixer + Multiscale patches	79.5	8.1	0.884	4	0.343
Spectral Mixer + Scale-space bands	78.5	8.8	0.855	3	0.352
Spectral Mixer + Gradient + shape	76.2	9.6	0.858	1	0.345
Patch Transformer + Gradient + shape	75.9	8.4	0.819	3	0.746
Patch Transformer + Scale-space bands	75.2	7.9	0.790	0	0.747
ModernTCN-lite + Gradient + shape	75.0	5.8	0.765	1	8.693
ConvNeXt-1D + Gradient + shape	74.9	4.8	0.799	1	3.518
Spectral Mixer + Frequency hybrid	74.2	8.6	0.841	0	0.346
ConvNeXt-1D + Scale-space bands	74.1	6.1	0.774	0	3.523
ModernTCN-lite + Scale-space bands	73.6	6.4	0.791	0	8.638
ModernTCN-lite + Frequency hybrid	73.6	5.6	0.744	0	8.673
ModernTCN-lite + Multiscale patches	73.1	6.3	0.766	0	8.579
Patch Transformer + Frequency hybrid	71.5	8.1	0.780	0	0.761
ConvNeXt-1D + SNV raw	71.4	7.1	0.768	0	3.530
ModernTCN-lite + SNV raw	71.4	8.4	0.757	0	8.713
Patch Transformer + Multiscale patches	71.1	7.1	0.748	0	0.736
ConvNeXt-1D + Multiscale patches	70.6	6.5	0.773	0	3.526
ConvNeXt-1D + Frequency hybrid	66.9	7.8	0.767	1	3.503

Note: Reported AUC is the pooled inner-CV value, not an outer-test metric. The mean and SD for balanced accuracy are expressed in percent and percentage points.

Figure 2 | Joint classifier × representation ablation

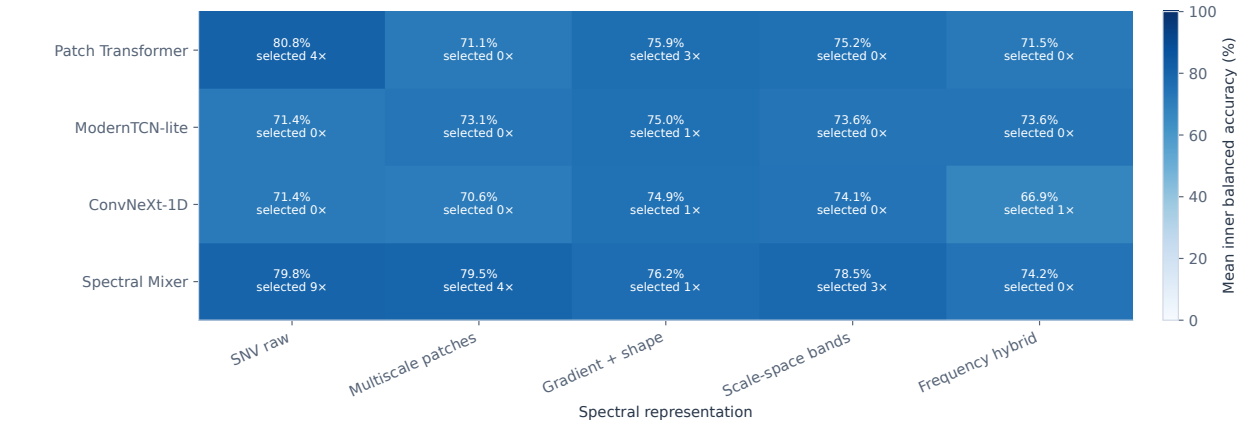


Figure 2: Joint classifier–representation ablation. Cell annotations show mean inner-CV balanced accuracy and how many outer folds selected each configuration; selection frequency is not test accuracy.

1.4 Outer evaluation: Does augmentation improve predictions on original patients?

Table 3 and Fig. 3 report the patient-level outer results. The most informative comparison for augmentation is between *matched pair: real only* and *matched pair: best synthetic*. In each outer fold these arms use the same selected classifier and representation; their difference is the inclusion of synthetic data during training. The real-only matched pipeline classified 22 of 27 patients correctly (81.5% accuracy) and obtained 73.6% BA. The corresponding augmented pipeline classified 23 patients correctly (85.2% accuracy) and obtained 80.7% BA, giving observed increases of 3.7 pp in accuracy and 7.1 pp in BA.

The confusion matrices explain the nature of the change. Both matched pipelines correctly classified 18 of 20 sick patients, corresponding to 90.0% sensitivity and two false negatives. The real-only pipeline correctly classified four of seven healthy patients, whereas the augmented pipeline classified five of seven healthy patients correctly. Consequently, specificity increased from 57.1% to 71.4%, while sensitivity was unchanged. The F1 score increased from 0.878 to 0.900 and MCC from 0.497 to 0.614. In this dataset, the observed BA improvement was driven by the healthy class, which had only seven members; changing a single healthy classification alters specificity by approximately 14.3 pp and BA by approximately 7.1 pp.

The final policy that selected real-only or synthetic training through inner CV produced the same aggregate hard predictions as the matched best-synthetic arm: 23/27 correct, 80.7% BA, 90.0% sensitivity, and 71.4% specificity. However, the reported average precision differed slightly (0.953 versus 0.955), so identical classification decisions do not imply identical probability scores.

Among the architecture-specific real-only baselines, Patch Transformer achieved the highest BA (76.1%) and 23/27 correct predictions. Its sensitivity was 95.0%, but specificity was 57.1%. ConvNeXt-1D obtained 68.2% BA (18/27 correct), ModernTCN-lite obtained 58.6% (16/27), and Spectral Mixer obtained 64.3% (22/27). Notably, the real-only Spectral Mixer correctly identified all 20 sick patients, but identified only two of the seven healthy patients (28.6% specificity). Its high raw accuracy of 81.5% therefore concealed an imbalance between class-wise outcomes.

The augmented pipeline had the largest observed BA in the benchmark, but comparisons with separately configured real-only architectures involve differences in both modeling choices and training strategy. Such comparisons are useful descriptive benchmarks, whereas the matched analysis more directly isolates augmentation.

Table 3: Patient-level evaluation on original outer-test samples ($N = 27$; 20 sick and 7 healthy patients). Panel A gives classification and discrimination metrics; Panel B gives additional summary metrics and confusion-matrix counts.

Panel A: Predictive performance

Pipeline	Correct	Acc. (%)	BA (%)	Sens. (%)	Spec. (%)	AUC	AP
Matched pair: best synthetic	23/27	85.2	80.7	90.0	71.4	0.886	0.955
CV-selected policy	23/27	85.2	80.7	90.0	71.4	0.886	0.953
Matched pair: real only	22/27	81.5	73.6	90.0	57.1	0.871	0.954
ConvNeXt-1D: real only	18/27	66.7	68.2	65.0	71.4	0.736	0.917
ModernTCN-lite: real only	16/27	59.3	58.6	60.0	57.1	0.686	0.871
Patch Transformer: real only	23/27	85.2	76.1	95.0	57.1	0.693	0.775
Spectral Mixer: real only	22/27	81.5	64.3	100.0	28.6	0.971	0.991

Panel B: Confusion matrices and complementary metrics

Pipeline	F1	MCC	TP	TN	FP	FN
Matched pair: best synthetic	0.900	0.614	18	5	2	2
CV-selected policy	0.900	0.614	18	5	2	2
Matched pair: real only	0.878	0.497	18	4	3	2
ConvNeXt-1D: real only	0.743	0.321	13	5	2	7
ModernTCN-lite: real only	0.686	0.151	12	4	3	8
Patch Transformer: real only	0.905	0.588	19	4	3	1
Spectral Mixer: real only	0.889	0.478	20	2	5	0

Note: Sick is the positive class. BA is balanced accuracy; AP, average precision; MCC, Matthews correlation coefficient. "Matched pair" holds the fold-specific classifier and representation fixed; only the use of augmentation changes. The remaining architecture baselines use real training data only. Probability scores used for pooled AUC/AP are not calibrated across outer folds.

Figure 3 | Patient-level benchmark: is augmentation beneficial?

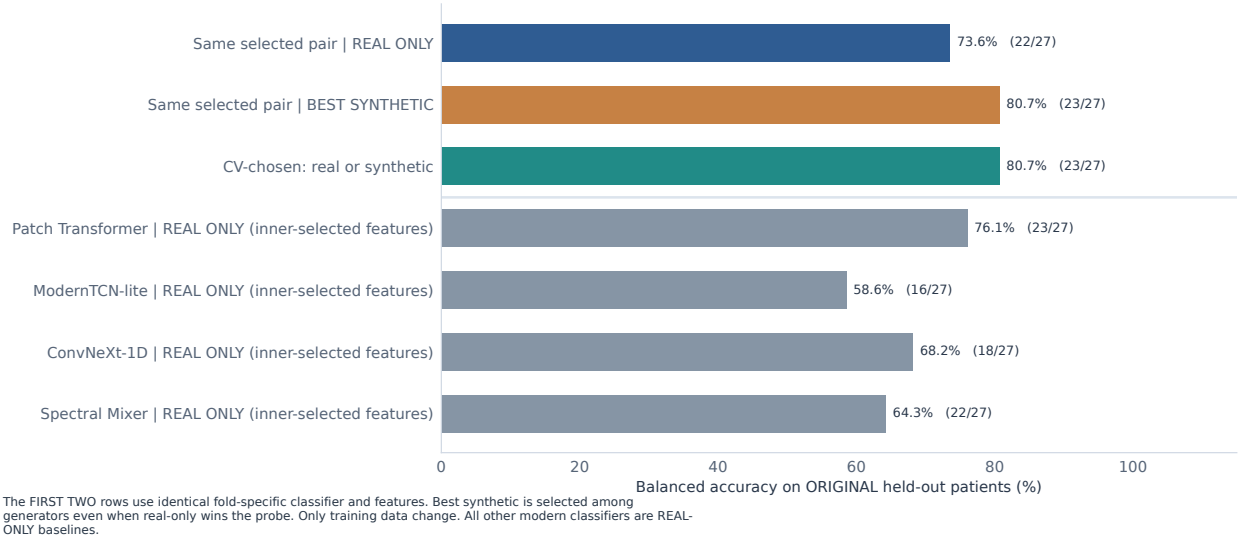


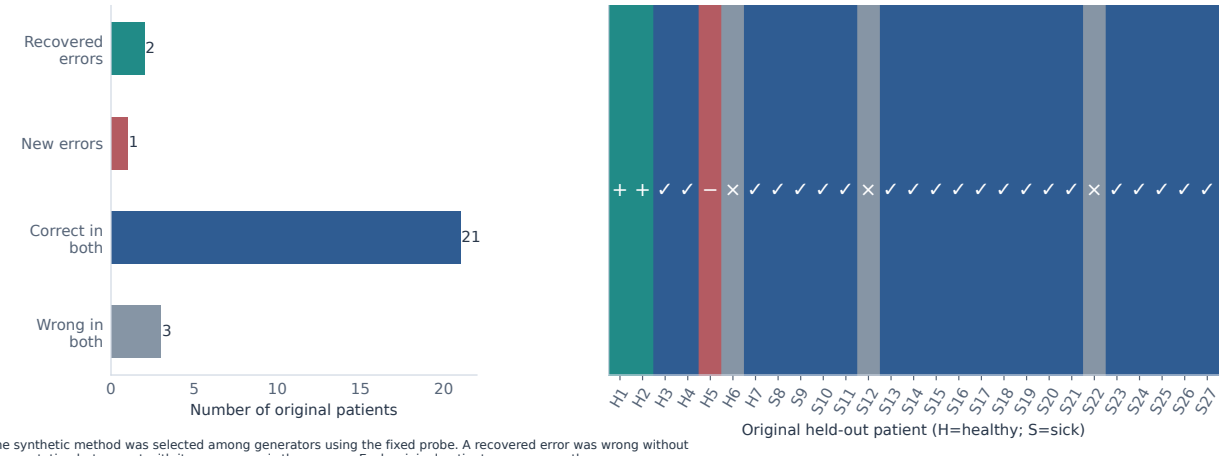
Figure 3: Balanced accuracy on 27 original held-out patients. The matched real-only and best-synthetic arms hold the selected classifier and representation constant, isolating synthetic training data.

1.5 Patient-level error transitions and clinical interpretation

Figure 4 compares the binary predictions of the matched real-only and matched best-synthetic pipelines for each original patient. Augmentation recovered two previously incorrect predictions and introduced one new error. The methods agreed on 21 correct classifications and three incorrect classifications. These four categories sum to 27 patients and explain why the total number of correct predictions rose from 22 to 23. The same aggregate sensitivity in both matched arms, together with an increase in true negatives from four to five, shows that the net improvement was attributable to the healthy subgroup rather than improved recognition of sick patients.

The error-transition analysis provides a more transparent interpretation than the BA difference alone. The augmented method did not improve every patient's classification: it corrected some decisions while reversing another correct decision. Given the small number of healthy patients, the practical relevance of the estimated specificity improvement cannot be established from this dataset alone. In particular, a gain in overall BA must not be interpreted as evidence of higher clinical sensitivity when the number of correctly identified sick patients did not change.

Figure 4 | Does augmentation correct errors or introduce new ones?



The synthetic method was selected among generators using the fixed probe. A recovered error was wrong without augmentation but correct with it; a new error is the reverse. Each original patient appears exactly once.

Figure 4: Paired error transitions for each original held-out patient. Recovered errors are corrected by augmentation; new errors result from previously correct predictions becoming incorrect.

1.6 Paired effect sizes and statistical uncertainty

Table 4 provides paired BA differences, exploratory 95% stratified-patient bootstrap intervals, paired error transitions, exact McNemar tests, and exact paired-BA permutation tests. Figure 5 shows the effect estimates and intervals graphically. The principal matched comparison gave $\Delta\text{BA} = +7.14$ pp, with a bootstrap interval from -14.29 to $+28.57$ pp. Because the interval crosses zero, the results are compatible with both an improvement and a deterioration. Both the exact McNemar test and the paired-BA permutation test returned $p = 1.000$ for this comparison, including after Holm correction.

Relative to the real-only architecture baselines, the best-synthetic pipeline achieved observed BA differences of +4.64 pp versus Patch Transformer, +12.50 pp versus ConvNeXt-1D, +16.43 pp versus Spectral Mixer, and +22.14 pp versus ModernTCN-lite. Nevertheless, every corresponding reported bootstrap interval included zero: $[-19.29, 28.57]$, $[-8.94, 34.29]$, $[-12.14, 42.50]$, and $[-4.29, 48.57]$ pp, respectively. These intervals indicate substantial uncertainty about the direction and magnitude of the generalization differences.

For ModernTCN-lite, the unadjusted exact McNemar test yielded $p = 0.039$, whereas the Holm-adjusted value was $p = 0.234$; the paired-BA permutation test yielded $p = 0.098$ before correction and $p = 0.586$ after correction. None of the multiple-comparison-adjusted tests supports a statistically established advantage at the 0.05 level. The CV-selected policy produced a zero hard-prediction

difference relative to the matched best-synthetic arm, consistent with the same recorded confusion matrices. The exploratory bootstrap intervals should also not be viewed as confirmatory evidence, particularly given the small healthy subgroup.

Accordingly, it is appropriate to report an *observed* improvement in matched held-out BA and specificity, but not to claim a statistically demonstrated benefit of generative augmentation or the superiority of one architecture across new patient populations.

Table 4: Paired comparisons of the best-synthetic pipeline against the CV-selected policy and real-only classifiers. Exploratory 95% intervals were estimated by stratified patient bootstrap.

Panel A: Balanced-accuracy effects and patient transitions

Comparator	ΔBA (pp)	95% CI (pp)	Recovered	New errors
CV-selected policy	+0.0	[+0.0, +0.0]	0	0
Matched pair: real only	+7.1	[−14.3, +28.6]	2	1
ConvNeXt-1D: real only	+12.5	[−8.9, +34.3]	6	1
ModernTCN-lite: real only	+22.1	[−4.3, +48.6]	8	1
Patch Transformer: real only	+4.6	[−19.3, +28.6]	2	2
Spectral Mixer: real only	+16.4	[−12.1, +42.5]	4	3

Panel B: Exact paired hypothesis tests

Comparator	McNemar p	McNemar–Holm p	Permutation p	Permutation–Holm p
CV-selected policy	1.000	1.000	1.000	1.000
Matched pair: real only	1.000	1.000	1.000	1.000
ConvNeXt-1D: real only	0.125	0.625	0.281	1.000
ModernTCN-lite: real only	0.039	0.234	0.098	0.586
Patch Transformer: real only	1.000	1.000	1.000	1.000
Spectral Mixer: real only	1.000	1.000	0.375	1.000

Note: ΔBA denotes augmented minus comparator in percentage points. "Recovered" means incorrect for comparator but correct for augmented; "new errors" means the reverse. Intervals are descriptive, and the Holm corrections address multiple comparisons. The CV-selected policy produced identical binary decisions to the best-synthetic pipeline.

Figure 5 | Size and uncertainty of the augmentation benefit

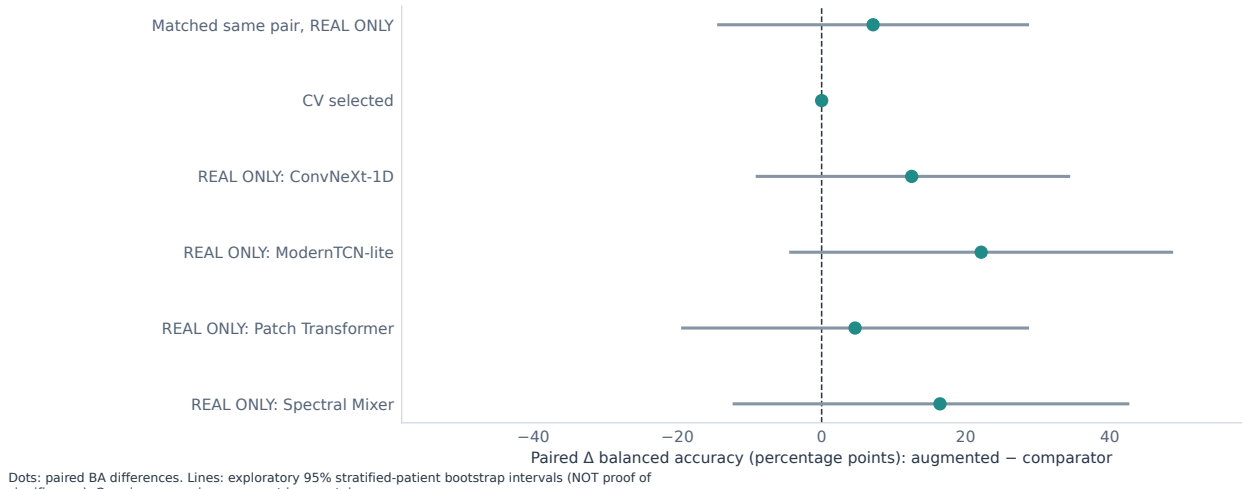


Figure 5: Paired differences in balanced accuracy, with exploratory stratified-patient bootstrap 95% intervals. Confidence intervals spanning zero indicate uncertainty in the direction of the effect.

1.7 Threshold-independent discrimination

The receiver-operating-characteristic (ROC) and precision–recall (PR) curves in Fig. 6 examine discrimination among the original held-out patients. The matched real-only pipeline yielded an AUC of 0.871 and an AP of 0.954. With best-synthetic augmentation, AUC increased to 0.886 and AP to 0.955. The final CV-selected policy also produced an AUC of 0.886, with an AP of 0.953. These small differences indicate that the classification-threshold gain in BA was not accompanied by a comparably large change in the reported ranking summaries.

The distinction between discrimination and binary decisions is particularly visible for the real-only Spectral Mixer in Table 3: its AUC was 0.971 and AP was 0.991, while its BA at the evaluated operating threshold was only 64.3%. This combination is mathematically possible because AUC and AP describe how scores rank patients over varying thresholds, whereas BA depends on a particular threshold and the resulting confusion matrix. It also emphasizes that threshold-dependent specificity matters when healthy patients are the minority.

The ROC and PR curves pool predictions from separately fitted outer-fold models. Their scores were not calibrated onto a common cross-fold scale, so the pooled AUC and AP should be treated as descriptive secondary analyses rather than definitive estimates of discrimination for one deployed model. The patient-level confusion matrices and paired BA differences are better aligned with the primary evaluation objective. With 20 sick patients among 27, the positive-class prevalence is 74.1%, which should also be borne in mind when interpreting high AP values.

Figure 6 | Discrimination among held-out original patients

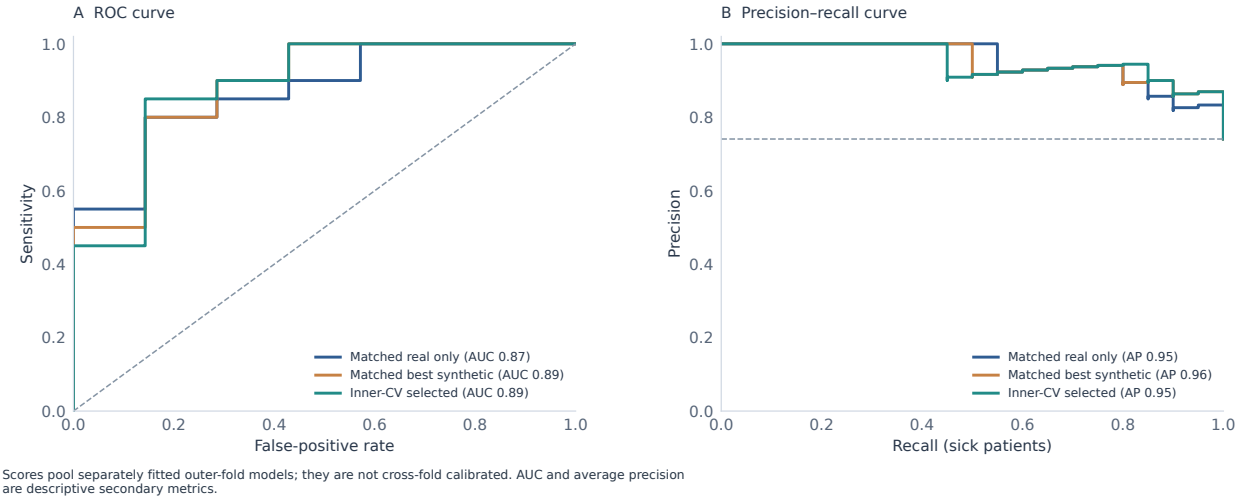


Figure 6: Pooled outer-fold ROC (left) and precision–recall (right) curves for the matched and CV-selected pipelines. Scores are not calibrated across outer folds; reported AUC and AP are descriptive.

1.8 Visual fidelity of synthetic spectral profiles

Figure 7 examines synthetic-data plausibility for VP Flow Matching by comparing the real-training and generated distributions for healthy and sick spectra within one outer training fold. The curves present median SNV-normalized intensity across wavenumber, together with 10th–90th percentile bands. The main intensity peaks occupy similar spectral positions in the real and generated medians in both classes, while several smaller peaks and oscillatory structures are also reproduced. The band overlays suggest that the generator captures part of the amplitude variability seen in the training spectra, although local discrepancies are visible.

These observations support only a qualitative statement about spectral resemblance in the illustrated fold. Visual overlap of median traces and quantile bands does not verify patient-level diversity, the preservation of disease-specific molecular information, the absence of memorization, or the utility of the generated signals for an independent patient cohort. Furthermore, the visual

comparison displays VP Flow Matching rather than the Rectified Flow configuration most frequently selected in the first-stage screening. It therefore cannot be used as evidence that the selected generator is more biologically plausible than the alternatives.

Figure 7 | Spectral plausibility of synthetic data (VP Flow Matching)

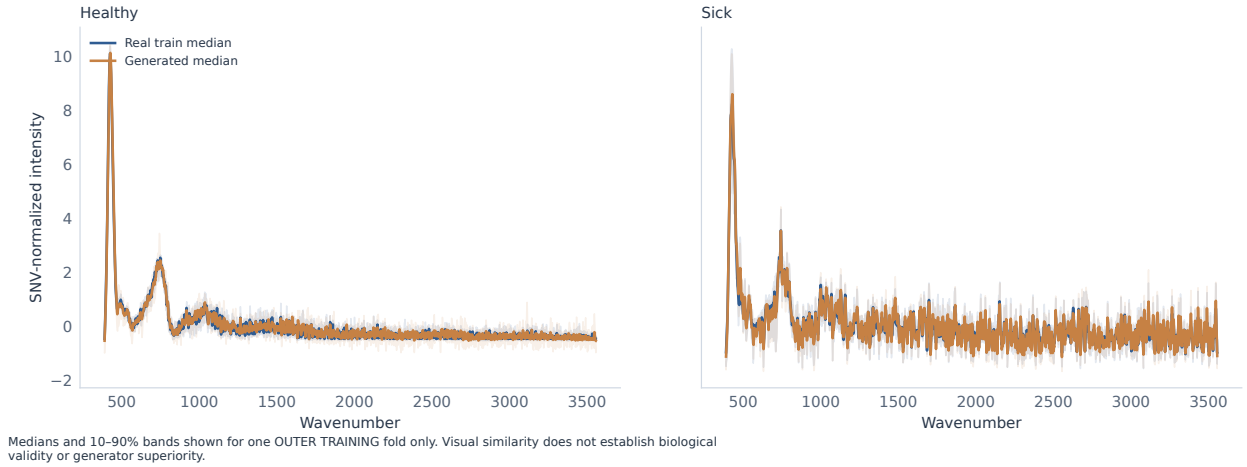


Figure 7: Real-training and VP Flow Matching-generated SNV-normalized spectra in one outer training fold, shown as median traces and 10th–90th percentile bands separately for healthy and sick classes.

1.9 Accuracy–computation trade-offs

The computational comparison in Fig. 8 positions each inner-search candidate by validation BA and reported mean cold-start training time, with the selected candidate indicated for each stage. In Stage 1, real-only training required 0.225 s per inner fold, while Rectified Flow with 12 synthetic spectra per class required 0.612 s. Thus, the mean time increased by 0.387 s (approximately 2.7-fold) for an observed BA gain of roughly 2.3–2.4 pp. Using 30 samples per class further increased the reported mean time for Rectified Flow to 1.020 s, without improving its BA relative to the 12-sample configuration.

In Stage 2, Spectral Mixer with raw SNV spectra required 0.342 s per inner fold and achieved 79.8% BA. Patch Transformer with raw SNV achieved the highest mean BA of 80.8% but required 0.742 s, approximately 2.2 times the Spectral Mixer time. For comparison, the raw-SNV ConvNeXt-1D and ModernTCN-lite configurations required 3.530 and 8.713 s, respectively, while attaining only 71.4% BA each. These measurements indicate that the Spectral Mixer with raw SNV spectra provided a favorable trade-off among the evaluated candidates, explaining its frequent fold-specific selection despite not attaining the highest mean BA.

All reported times refer to the measured inner-fold cold-start evaluation and, according to the figure, include generator fitting. They are not end-to-end deployment latencies, and no timing-based claim is made for alternative hardware, dataset sizes, or training implementations.

Figure 8 | Is extra computation justified by better validation accuracy?

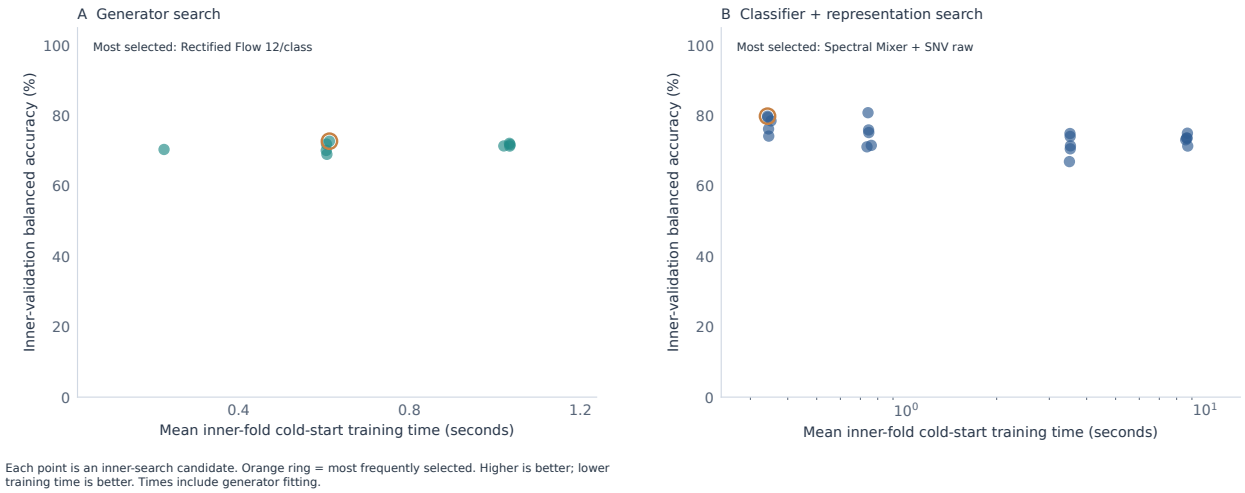


Figure 8: Trade-off between mean inner-CV balanced accuracy and reported cold-start training time for augmentation screening (left) and classifier–representation selection (right). Circled points mark the most frequently selected candidates.

1.10 Synthesis and limitations

The experiments provide three related findings. First, generative augmentation changed inner-CV performance in a generator- and budget-dependent manner; the highest-scoring generator improved the fixed-probe BA only modestly, and no single generator was consistently selected. Second, model choice interacted with spectral representation: Patch Transformer with raw SNV achieved the highest mean inner-CV BA, while Spectral Mixer with raw SNV was selected more frequently and required less computation. Third, augmenting the fold-specific selected classifier–representation pair increased observed outer BA from 73.6% to 80.7%, with unchanged sensitivity and improved specificity. This apparent gain amounted to one additional correct classification across 27 held-out patients and was not supported by the corrected paired tests.

Several limitations constrain the interpretation. The available benchmark contains only seven healthy patients, making estimates of specificity and BA unstable; one healthy prediction alone changes BA by about 7.1 pp. The nested procedure protects the outer test set from direct use in configuration selection, but it does not replace validation on a separate external cohort. The number of models and augmentation strategies is relatively large compared with the patient sample, and fold-specific selection varies. Additionally, the pooled discrimination scores are not cross-fold calibrated, and one-fold spectral overlays cannot establish biological fidelity. The current evidence therefore supports feasibility and a descriptive augmentation benefit on this dataset, rather than validated diagnostic superiority. A larger, independently collected patient cohort, replicated generator training, and prospective assessment of class-specific errors are required before stronger generalization claims are justified.